Shelf Voice — 01 Product Requirements

Oct 27, 2026 · @Aradhya Chaudhary

Status and open decisions

Version 1.0, approved by the owner on Oct 27. All six decisions are final; changes from here go through the decision log.

#	Decision	Options	Recommendation	Owner's choice
D1	Hackathon track and team size	Indie solo; Indie team of 2-4; Startup track	Indie track (Startup needs an existing startup with an employee as Representative)	Indie track, team of 3
D2	Primary platform	Mobile web app (installable PWA); Native Android app; WhatsApp bot	PWA, 70%: mic and camera work in the browser, no app-store review, judges open one link	PWA
D3	Demo languages	Hindi + English + Hinglish; plus one South Indian language; English only	Hindi + English + Hinglish, 65%: matches the core user; all validated for Muse Voice Transcribe	Hindi + English + Hinglish
D4	What customers can do on the storefront	Browse + send order on WhatsApp; Browse only; Full in-app cart and orders	WhatsApp order, 60%: matches how kirana customers already order; no payments to build	Browse + order via WhatsApp
D5	How the owner logs sales and stock	Voice + bill photo; Voice only; Import from a billing app or UPI export	Voice + bill photo, 60%: shows two Meta modalities, needs no third-party access	Voice + bill photo
D6	Customer feedback channel	Web voice recorder opened by QR; WhatsApp voice note	Web recorder, 75%: no WhatsApp Business API approval needed within 8 days	Web recorder via QR

Product overview

Shelf Voice is a voice-first assistant that turns how a small shop owner already talks about their shop into a live storefront, a restock forecast and a customer feedback loop, all running on one shared catalog.

One-line pitch: speak your shop online, know what to restock, and hear every customer, without typing.

It is built for the Meta Global AI Developer Hackathon (Oct 26 to Nov 4, 2026) on the Meta Model API: Muse Spark for reasoning and structured output, Muse Voice Transcribe for speech, and SAM 3.1 for product images.

Problem statement

Small retail shop owners, especially India's roughly 13 million kirana stores, run their business on memory and guesswork, because every digital tool built for them assumes typing, design skills and data literacy.

That single root cause shows up as three problems:

Symptom	What happens today	Source problem (ITCH score)
Invisible online	Website builders need design and content-structure skills, so products, hours and contact details never get published	Voice-to-website for shop owners (92.4)
Restocking by guesswork	Fast movers run out, slow sellers pile up, because affordable demand forecasting does not exist at this scale	Kirana restock prediction (61.5)
Silent customers	Long surveys go unanswered, so complaints like "you never have atta" never reach the owner	Quick voice feedback (70.8)

All three are missing the same asset: a structured record of what the shop sells, what actually sells, and what customers ask for. Shelf Voice builds that record from speech and photos, then uses it three ways.

Target users and personas

The primary user is the shop owner; customers are a secondary user who never needs an account.

Persona	Who	Goals	Pain points	Device and skills
Ramesh, kirana owner (primary)	45, runs a 300-SKU grocery store in a tier-2 city, works the counter 12 hours a day	Get more neighbourhood orders, stop losing sales to stock-outs, keep regulars happy	No time or skill for websites; orders stock from memory; hears complaints only when a regular stops coming	Android phone, uses WhatsApp and UPI daily, speaks Hindi-English mix, types slowly
Priya, neighbourhood customer (secondary)	30, orders groceries from nearby shops and quick-commerce apps	Check if the shop has an item and its hours before walking over; order quickly	No way to see the shop's stock or hours online; feedback has no channel	Smartphone, WhatsApp, scans QR codes routinely
Hackathon judge (evaluator)	Reviews the demo video and live product	See real impact, creative Meta model use, sound engineering, a demo that holds up live	Many projects are thin chat wrappers	Laptop, opens a link, watches a 2-3 minute video

Assumption: the owner has an Android phone with a working mic, camera and mobile data. Feature phones are out of scope.

Value proposition and why now

For the owner, Shelf Voice replaces three tools they would never set up with one they only have to talk to.

- Online in minutes: a 60-second spoken description becomes a published storefront.
- Fewer stock-outs: plain-language restock advice ("order 2 cases of Maggi before Friday") instead of charts.
- Customer voice: 5-second voice reviews that land as sentiment and demand signals, not a pile of audio.

- One loop: every sale, restock and review improves the same catalog, so each feature makes the others better.

Why now:

- Speech recognition now handles Indian code-switching. Muse Voice Transcribe launched with Hindi, Marathi, Bengali, Tamil, Telugu and Kannada among 25 validated languages, and transcribes mixed-language speech natively.
- Quick-commerce apps are pulling sales away from kiranas, and many stores still have no digital presence, so the need is rising.
- Running costs are low enough for a micro-business: transcription is listed at about $0.18 per audio hour on OpenRouter.

Business objectives and success metrics

The objective for this hackathon is a working, demo-proof product that scores on all four judging criteria, validated with real shop owners.

Objective	Judging criterion (weight)	Metric	Target
Owners go online without typing	Impact (30%)	Time from first voice note to published storefront	Under 3 minutes for a 15-product shop
Accurate catalog from speech	Technical depth (20%)	Products captured with correct name, quantity and price, on a 30-utterance Hindi/Hinglish test set	At least 90%
Useful restock advice	Impact (30%)	Backtest on 90 days of seeded sales: stock-outs flagged at least 2 days ahead	At least 80%
Effortless customer feedback	Demo (25%)	Time from QR scan to submitted voice review	Under 15 seconds
Reliable feedback insight	Technical depth (20%)	Sentiment and item tags agreeing with human labels on 30 recorded reviews	At least 85%

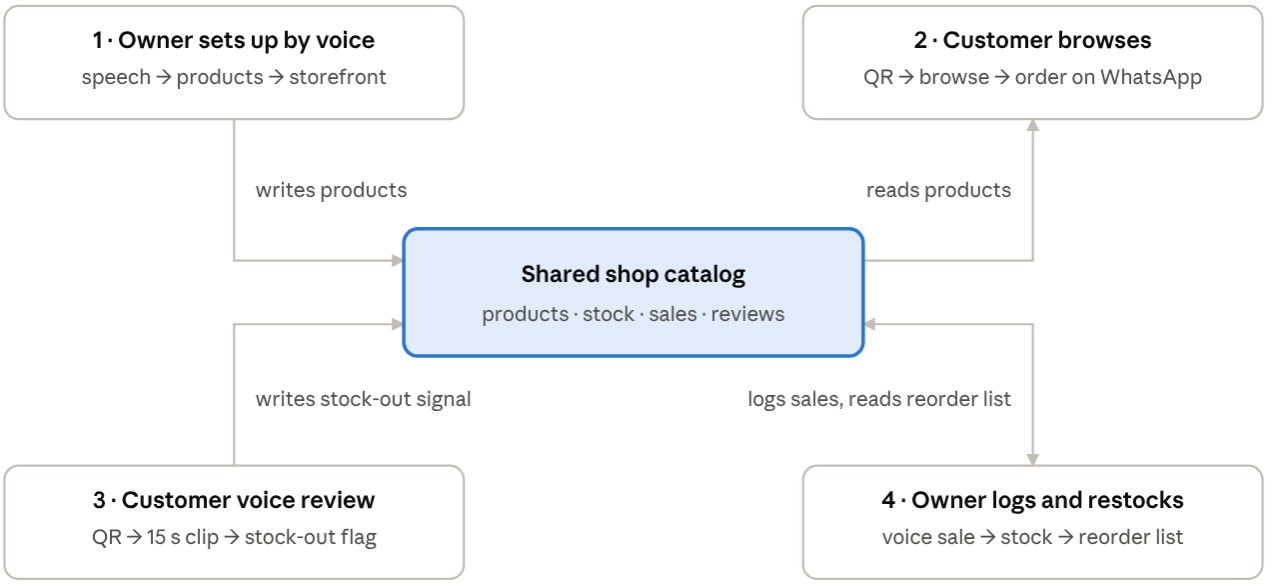
Objective	Judging criterion (weight)	Metric	Target
Creative Meta model use	Meta tooling (25%)	Distinct Meta models doing real work in the core flow	3 (Muse Spark, Muse Voice Transcribe, SAM 3.1)
Real-world validation	Impact (30%)	Real shop owners who try the storefront flow, with quotes for the video	3
Stays within credits	Constraint	Total Meta API spend through submission	Under $100 of the $150 credit

Targets are proposals; the test sets and backtest method are defined in document 04 (Testing and Verification).

Core user journeys

Four journeys make up the product, and each one either writes to or reads from the same shop catalog; that loop is the demo story.

Every journey reads or writes one shared shop catalog



core journeys · 4 journeys around one catalog

1. Owner sets up by voice: sign in, speak about the shop for about a minute, review the extracted products, publish, print the QR.

2. Customer browses: scan the QR or open the link, check items and hours, send an order on WhatsApp (per D4).
3. Customer voice review: scan the counter QR, record up to 15 seconds, done; the owner sees transcript, sentiment and any stock-out flag.
4. Owner logs and restocks: speak sales and deliveries during the day; open the restock list to see what to order, how much, and why.

Functional requirements

Ten Must requirements make up the demo loop; Should items are built only after every Must passes its acceptance criteria.

ID	Requirement	Priority	Meta model doing the work
FR-01	Owner signs in with Google (changed from phone OTP on Oct 27, Technical decision T4, because Indian SMS OTP needs DLT registration); one owner account owns one shop; judges get a one-tap demo sign-in	Must	None
FR-02	Voice shop setup: owner speaks about the shop; the system extracts shop profile (name, type, hours, address, phone) and a product list (name, unit, price, category)	Must	Voice Transcribe, Muse Spark (structured output)
FR-03	Review screen: owner sees the extracted profile and products, fixes anything by tap or by voice, then publishes	Must	Muse Spark
FR-04	Public storefront: shareable link and printable QR, products by category with price and in-stock badge, hours with open/closed now, address, call and WhatsApp buttons	Must	None

ID	Requirement	Priority	Meta model doing the work
FR-05	Voice catalog edits: add, remove, re-price or mark out of stock by speaking ("Maggi ka price 15 kar do")	Must	Voice Transcribe, Muse Spark (tool calling)
FR-06	Voice sales and stock logging: "aaj 12 packet Maggi gaye" updates stock and the sales ledger	Must	Voice Transcribe, Muse Spark (tool calling)
FR-07	Restock view: per item, days of stock left, reorder quantity, and a one-line reason in the owner's language	Must	Muse Spark (explanation only; numbers computed in code)
FR-08	Customer voice review: QR opens a recorder, customer records 5-15 seconds, no account needed	Must	Voice Transcribe
FR-09	Review insight: each review gets sentiment, mentioned items and a stock-out flag; flagged items rise in restock priority	Must	Muse Spark
FR-10	Shelf photo to products: segment items on a shelf photo, name them, and use the cut-outs as storefront images	Should	SAM 3.1, Muse Spark (vision)
FR-11	Supplier bill photo to restock entry: read line items and add them to stock	Must (per D5)	Muse Spark (vision)
FR-12	Festival and season bump: raise forecasts ahead of known festivals	Should	None (calendar rule)
FR-13	Daily summary card: today's sales, items running low, new reviews	Should	Muse Spark
FR-14	Storefront in the customer's choice of Hindi or English	Could	Muse Spark (translation)
FR-15	Storefront view counts for the owner	Could	None

Added Oct 27 from Security decision P6: FR-16 (Must) — the owner can delete their shop from settings, removing all of its data and files within 24 hours. Customer voice clips are kept 30 days then deleted (P1), so story 6's audio playback covers the last 30 days.

User stories and acceptance criteria

Each Must requirement has one story; a story is done only when every criterion under it is verified.

1. As an owner, I want to describe my shop by speaking, so that I get a storefront without typing. (FR-02, FR-03, FR-04)
   - Given a 60-second Hindi/Hinglish voice note naming 15 products with prices, at least 13 appear correctly on the review screen.
   - Hours, phone and address mentioned in speech are filled in; anything not mentioned is left blank and marked for the owner, never invented.
   - After Publish, the storefront loads on a phone at its public link in under 3 seconds on a 4G connection.
2. As an owner, I want to change my catalog by voice, so that updates take seconds. (FR-05)
   - "Maggi ka price 15 kar do" changes only Maggi's price and shows a confirmation the owner can undo.
   - If the item is ambiguous or not found, the app asks which item instead of guessing.
3. As an owner, I want to log sales and deliveries by voice, so that stock stays current without a register. (FR-06)
   - A sale reduces stock and adds a sales record with the date; a delivery increases stock.
   - Stock never goes below zero silently; the owner is asked to correct the count.
4. As an owner, I want to know what to reorder and why, so that I stop running out of fast sellers. (FR-07, FR-09)
   - Each low-stock item shows days left, a suggested quantity and a one-line reason in the owner's chosen language.
   - An item flagged as out of stock in a customer review moves up the list within one minute.
   - Suggested quantities match the forecasting formula in document 02 exactly; the model only writes the explanation.
5. As a customer, I want to leave a quick voice review, so that the shop hears me without filling a survey. (FR-08)

- Scanning the QR opens the recorder in one tap, with no sign-up; recording stops at 15 seconds.
- The customer sees a thank-you within 5 seconds of submitting, even if analysis is still running.

6. As an owner, I want reviews summarised for me, so that I act on them without listening to every clip. (FR-09)
   - Each review shows transcript, sentiment, items mentioned and a stock-out flag; the original audio stays playable.

Non-functional requirements

The product must feel instant on a mid-range Android phone and never lose an owner's data.

Area	Requirement
Performance	Storefront first load under 3 s on 4G; voice note to transcript under 5 s; transcript to extracted catalog under 20 s with a visible progress state
Devices	Works on Android Chrome 2022+ and desktop Chrome; layouts tested at 360, 768 and 1280 px wide
Language	Owner interface and AI replies in Hindi, English or Hinglish (per D3); speech in any mix of these
Accessibility	WCAG 2.1 AA for contrast, labels and keyboard use; large tap targets for one-handed counter use
Reliability	No data loss on a dropped connection: recordings retry upload; every AI change is confirmed before it is saved
Security and privacy	Owner data isolated per shop at the database level; API keys only on the server; customer reviews collected without names or phone numbers (details in document 03)
AI correctness	The model never invents prices, stock counts or reorder quantities; numbers come from the owner or from code
Cost	Meta API spend tracked per request; demo stays under $100 of the $150 credit

Area	Requirement
Observability	Every AI call logged with latency, token count and outcome, without raw customer audio in logs (document 05)

MVP scope and exclusions

The MVP is the full loop at demo quality: storefront polished, logging and restock complete but simple, feedback complete but simple.

In scope for submission (Nov 4):

- All Must requirements: FR-01 to FR-09, plus FR-11 (bill photo).
- FR-10 (shelf photo with SAM 3.1) if Must items pass by Nov 1, because it adds a third Meta model to the demo.
- One seeded demo shop with 90 days of realistic sales history, so the forecast has data on camera.
- A 2-3 minute demo video and a live demo link.

Explicitly excluded (Won't, this hackathon):

Excluded	Why
Online payments and checkout	Payment compliance and testing do not fit 8 days; customers order via WhatsApp or in person
Delivery and rider tracking	A separate product
Supplier ordering and price comparison	Needs supplier integrations that do not exist
Barcode scanning and POS hardware	Contradicts the voice-first premise and adds hardware
GST invoicing and accounting	Regulated domain; out of the core problem
Multi-store chains and staff accounts	One owner, one shop keeps permissions simple
Native mobile app and offline mode	Depends on D2; a PWA covers the demo
Trained ML forecasting model	Too little data per shop; a transparent formula is more honest and testable

Constraints, assumptions and dependencies

The hard constraint is time: submission closes Nov 4, 9:00 PM ET, which is Nov 5, 7:30 AM IST.

Constraints:

- About 8 days of build time from Oct 27, with a team of 3.
- $150 of Meta Model API credit, valid for the 10-day window.
- Rules require the project to be our original work; open-source components are allowed under their licenses.
- Anything submitted is non-confidential and licensed to the organisers, so no proprietary code or private data goes in the repo or demo.

Assumptions (to verify on day 1):

- The hackathon API key works from India. Meta's launch post described the public preview as US-only.
- Muse Voice Transcribe handles Hinglish counter speech, including product names like "Parle-G" and "Surf Excel", with keyword biasing.
- SAM 3.1 is callable through the Meta Model API with the same key.
- Meta does not train on our prompts at the standard tier; the cheaper contributor tier is not used.

Dependencies:

Dependency	Used for	Status
Meta Model API (Muse Spark)	Extraction, tool calling, explanations	Key to be tested
Muse Voice Transcribe	Owner and customer speech	Key to be tested
SAM 3.1	Product cut-outs from shelf photos	Availability to be confirmed
Database, auth, hosting	Chosen in document 02	Not chosen
Google sign-in (Supabase Auth)	Owner sign-in	Chosen in document 02 (T4)

Risks and mitigations

The two risks that could sink the demo are API access from India and Hinglish accuracy; both are tested on day 1, before any feature work.

Risk	Likelihood	Impact	Mitigation
API key does not work from India	Medium	Critical	Test on day 1; ask organisers on Discord; fall back to OpenRouter, which also serves Muse Spark and Muse Voice Transcribe
Hinglish product names transcribed wrongly	Medium	High	Keyword biasing with the shop's own catalog; review screen before anything saves; build a 30-clip test set early
Model invents products, prices or quantities	Medium	High	Strict structured output with schema validation; anything not spoken stays blank; all numbers in forecasts computed in code
Cold start: new shops have no sales history	Certain	Medium	Simple low-stock alerts from day 1; forecasts switch on after 14 days of logs; demo uses a seeded 90-day history, stated openly in the pitch
Scope too large for 8 days	High	High	Should items only after all Musts pass; daily checkpoint against the implementation plan (document 09)
Credits run out during testing	Low	High	Per-request cost tracking; cache repeated calls; small test sets
Abuse of the public review recorder (spam, offensive audio)	Medium	Medium	Rate limit per device and shop; length cap; owner can hide a review
Live demo fails on stage	Medium	High	Recorded backup video; seeded demo shop that resets on demand

Future opportunities

After the hackathon, the shared catalog and sales record open several paths; none are built now.

- Supplier connection: send the reorder list to the shop's distributor on WhatsApp in one tap.

- Neighbourhood discovery: customers search nearby Shelf Voice shops for an item that is in stock.
- More languages: Marathi, Bengali, Tamil, Telugu and Kannada are already validated for Muse Voice Transcribe.
- Offline and private mode: Muse Glimmer, Meta's open-weight 30B agentic model, could run on a shop-cluster server for owners who do not want data in the cloud. It needs about a 24 GB GPU, so it is not a phone option.
- Credit and working-capital signals: a clean sales history could support small-business lending, with the owner's consent.
- Voice replies: spoken answers to the owner once a suitable text-to-speech option is chosen.

Sources

Checked Oct 27, 2026. Pricing and rate figures from third-party pages are to be confirmed against Meta's own pricing page in document 02.

- Hackathon Official Rules and Event Waiver (provided by the owner): judging weights, tracks, credits, IP terms
- Hackathon deadline in UTC: Nov 4, 9:00 PM ET = Nov 5, 02:00 UTC
- Meta Model API product page: models available
- Build with Muse Spark on Meta Model API: OpenAI-compatible endpoint; preview described as US developers
- Muse Voice Transcribe, Hindi and Indian languages: code-switching, validated languages
- Muse Voice Transcribe supported language list
- Muse Voice Transcribe on OpenRouter: about $0.18 per audio hour
- Introducing Muse Glimmer
- Kirana stores turn to digitalisation: about 13 million stores, many without digital presence